

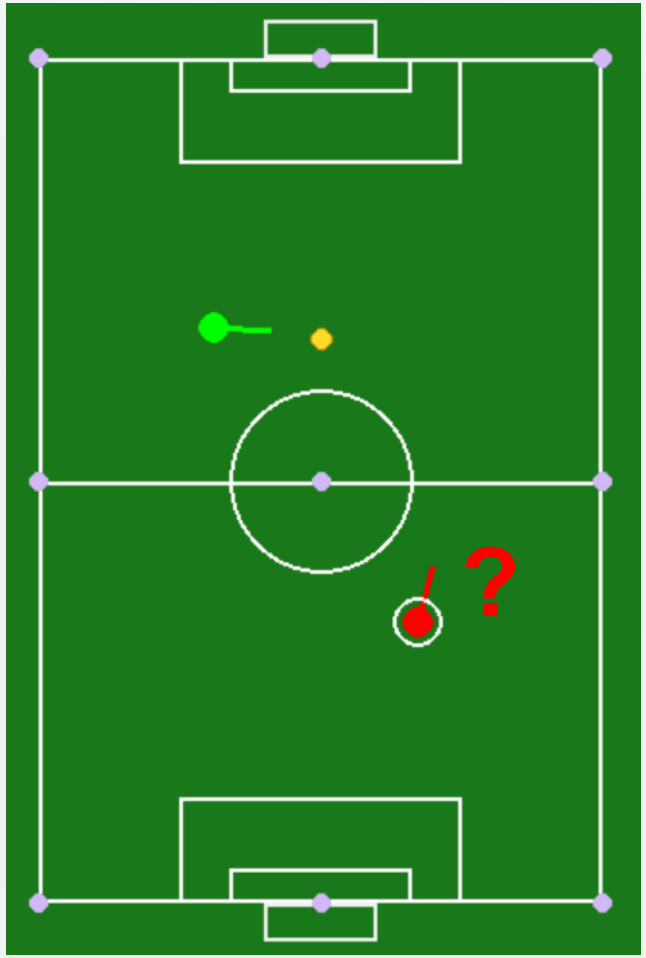
Robot Soccer Self-Localization with Active Inference

Robot football?



The idea

Where am I on the field?



Motion input

Velocity = $\begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix}_t$ =Odometry

Acceleration = $\begin{bmatrix} x'' \\ y'' \\ \theta'' \end{bmatrix}_t$ =IMU



Hidden state - location on the soccer field

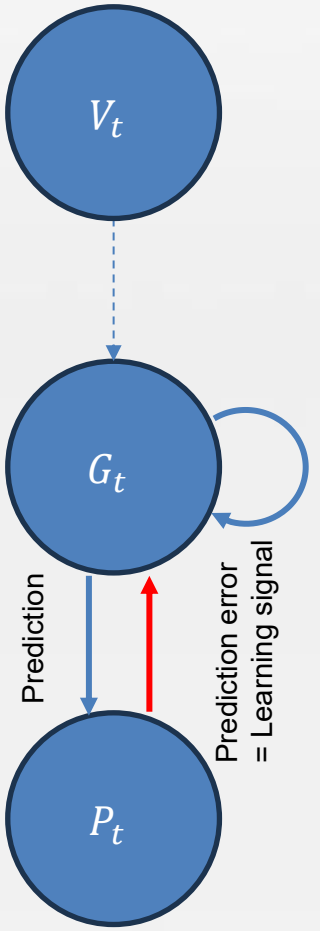
$\tilde{G}_t = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{\theta} \end{bmatrix}_t$ In generalized coordinates $\tilde{*} = \begin{bmatrix} * \\ *' \\ *'' \end{bmatrix} = \begin{bmatrix} position \\ speed\ at\ position \\ acceleration\ at\ position \end{bmatrix}$

Observed - landmarks

- predict landmarks
- Observation – prediction =

  sensory prediction error $\varepsilon_y = \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_\theta \end{bmatrix}$





Simulator results

Robocup requirements

- Fast-moving robots
  - reaching speeds up to 5 m/s and accelerations up to 12 m/s2
- Robust
  - no robots running out of the field
- Under noisy observations
  - including temporary obstruction
- ambiguous landmarks
- 110-degree field of view.
- Centimeter-scale tracking

Results

- efficient, robust, fast, and precise enough for robot soccer



Where will it be used

